EXP 13: V2X Latency Comparison Date

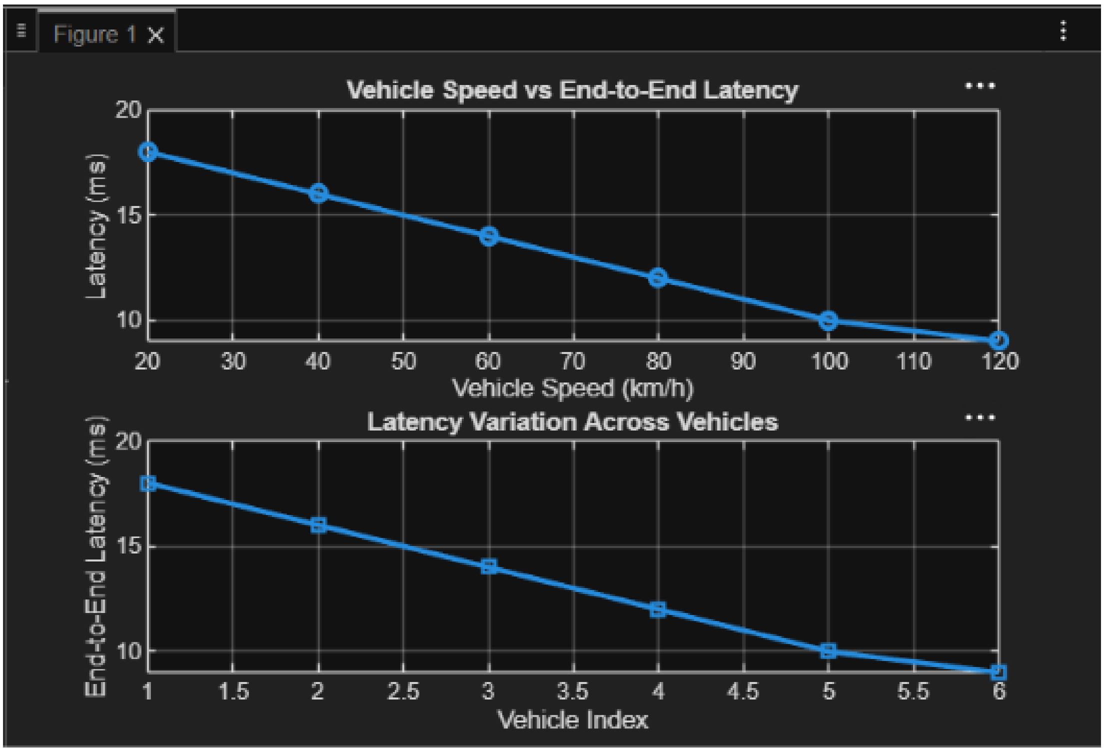
Objective 1 Matlab Code and Output

```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h) End-to-End Latency (ms)')
disp([speed' latency'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(1:length(latency),latency,'-s','LineWidth',2)
title('Latency Variation Across Vehicles')
xlabel('Vehicle Index')
ylabel('End-to-End Latency (ms)')
grid on
```

Output:

```
Command Window
Vehicle Speed (km/h) End-to-End Latency (ms)
    20    18
    40    16
    60    14
    80    12
   100    10
   120     9

>>
```



Objective 2 Matlab Code and Output:

Program code:

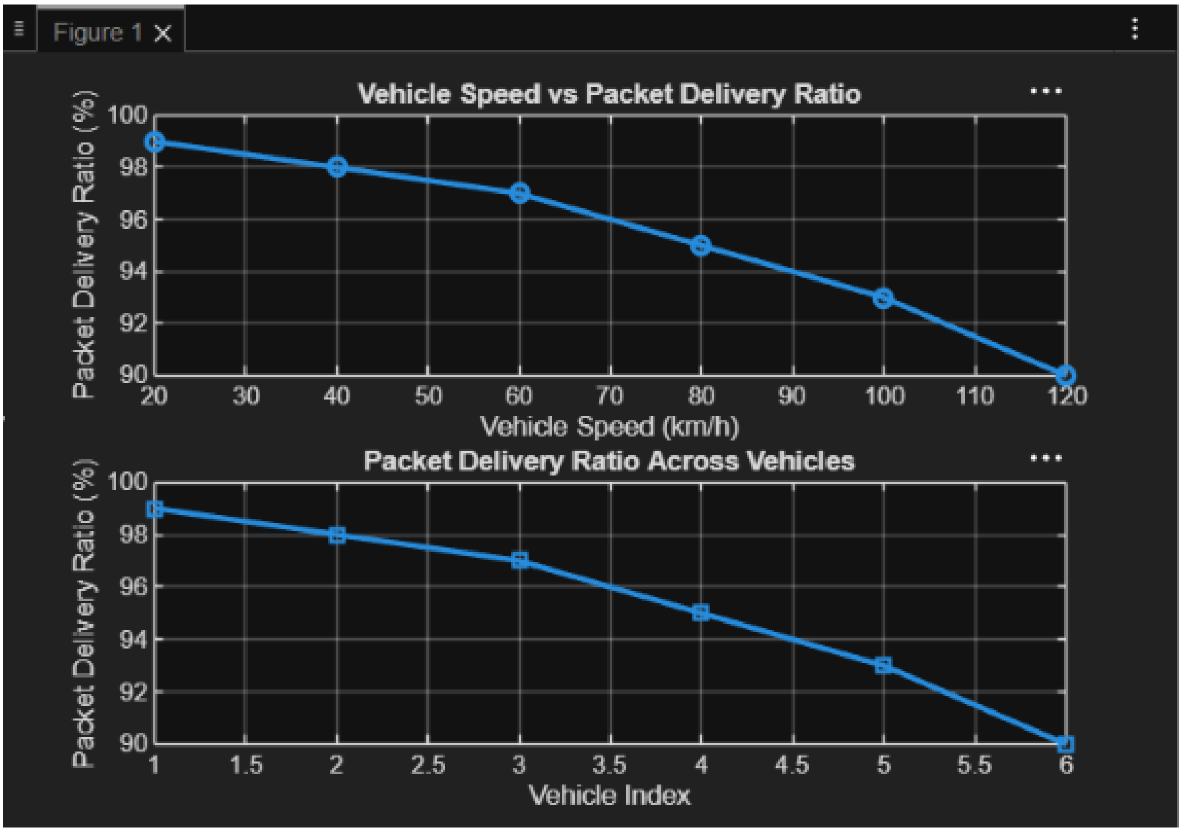
```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated Packet Delivery Ratio (%)
PDR = [99 98 97 95 93 90];
disp('Vehicle Speed (km/h) Packet Delivery Ratio (%)')
disp([speed' PDR'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(speed,PDR,'-o','LineWidth',2)
title('Vehicle Speed vs Packet Delivery Ratio')
xlabel('Vehicle Speed (km/h)')
ylabel('Packet Delivery Ratio (%)')
grid on

% -------- Graph 2 --------
subplot(2,1,2)
plot(1:length(PDR),PDR,'-s','LineWidth',2)
title('Packet Delivery Ratio Across Vehicles')
xlabel('Vehicle Index')
ylabel('Packet Delivery Ratio (%)')
grid on
```

Output:

```
Command Window
Vehicle Speed (km/h) Packet Delivery Ratio (%)
    20      99
    40      98
    60      97
    80      95
   100      93
   120      90

>>
```



Objective 3 Matlab Code and Output

Program code:

```
clc;
clear;
close all;
% Vehicle IDs
vehicle = 1:6;
% Vehicle Speed (km/h)
speed = [30 45 60 75 90 105];
% Communication Range (m)
range = [280 260 240 220 200 180];
```

```
disp('Vehicle Speed(km/h) Communication Range(m)')

disp([vehicle' speed' range'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(vehicle,speed,'-o','LineWidth',2)
title('Vehicle Speed of Different Vehicles')
xlabel('Vehicle Index')
ylabel('Vehicle Speed (km/h)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(speed,range,'-s','LineWidth',2)
title('Vehicle Speed vs Communication Range')
xlabel('Vehicle Speed (km/h)')
ylabel('Communication Range (m)')
grid on
```

Output:

```
Command Window
Vehicle Speed(km/h) Communication Range(m)
     1    30   280
     2    45   260
     3    60   240
     4    75   220
     5    90   200
     6   105   180

>>
```